



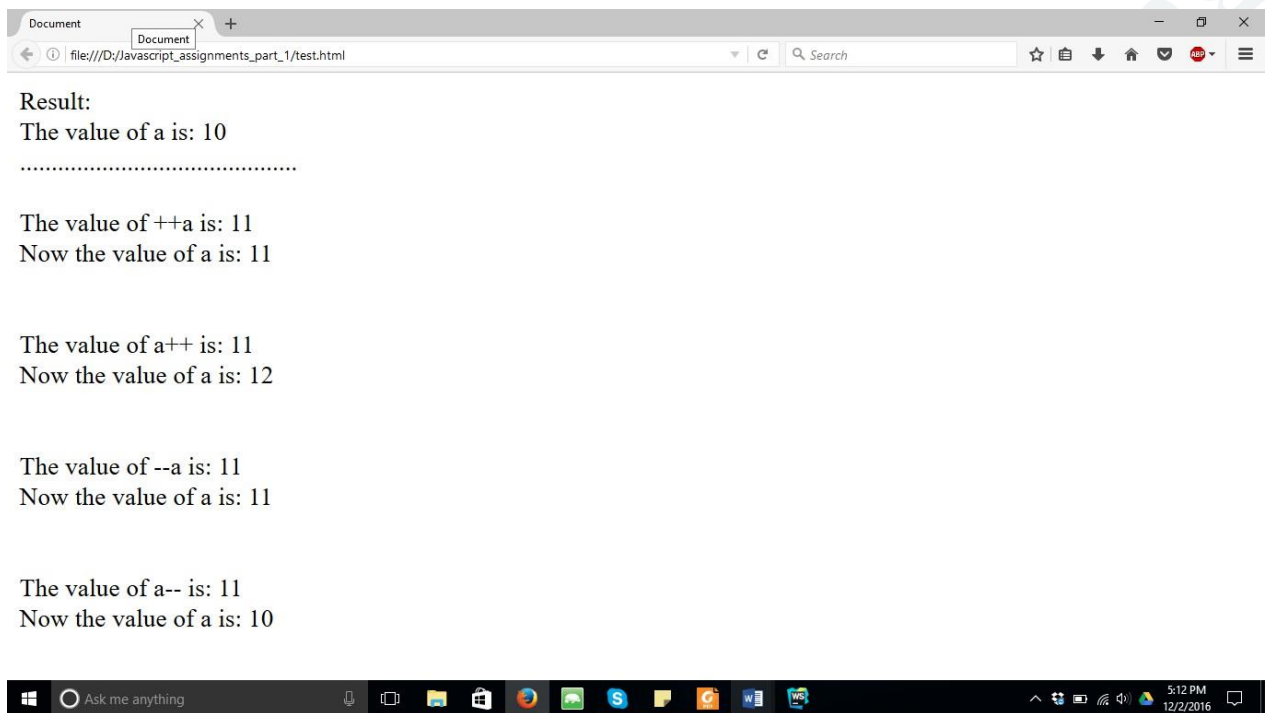
MATH EXPRESSIONS-

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Mobile & Cloud

Assignment # 6
JAVASCRIPT

1. Write a program to take a number in a variable, do the required arithmetic to display the following result in your browser:



CODE:

```
<!DOCTYPE html>

<html>

<head>

  <title>Question 1</title>

</head>

<body>


<h2>Result</h2>


<script>
```

```
let a = 10;

document.write("The value of a is: " + a + "<br><br>");

document.write("The value of ++a is: " + (++a) + "<br>");
document.write("Now the value of a is: " + a + "<br><br>");

document.write("The value of a++ is: " + (a++) + "<br>");
document.write("Now the value of a is: " + a + "<br><br>");

document.write("The value of --a is: " + (--a) + "<br>");
document.write("Now the value of a is: " + a + "<br><br>");

document.write("The value of a-- is: " + (a--) + "<br>");
document.write("Now the value of a is: " + a);

</script>

</body>

</html>
```

OUTPUT:

Result

The value of a is: 10

The value of ++a is: 11

Now the value of a is: 11

The value of a++ is: 11

Now the value of a is: 12

The value of --a is: 11

Now the value of a is: 11

The value of a-- is: 11

Now the value of a is: 10

2. What will be the output in variables ***a***, ***b*** & ***result*** after execution of the following script:

```
var a = 2, b = 1;
```

```
var result = --a - --b + ++b + b--;
```

Explain the output at each stage:

```
--a;
```

```
--a - --b;
```

```
--a - --b + ++b;
```

```
--a - --b + ++b + b--;
```

CODE:

```
<!DOCTYPE html>
<html>
<head>
  <title>Question 2</title>
</head>
<body>

<h2>Expression Result</h2>

<script>

let a = 2;
let b = 1;

let result = --a - --b + ++b + b--;

document.write("a = " + a + "<br>");
document.write("b = " + b + "<br>");
document.write("Result = " + result + "<br><br>");

document.write("<b>Explanation:</b><br><br>");

document.write("--a = 1<br>");
document.write("--a - --b = 1<br>");
document.write("--a - --b + ++b = 2<br>");
document.write("--a - --b + ++b + b-- = 3");

</script>

</body>
</html>
```

OUTPUT:

Expression Result

a = 1

b = 1

Result = 3

Explanation:

--a = 1

--a - --b = 1

--a - --b + ++b = 2

--a - --b + ++b + b-- = 3



3. Write a program that takes input a name from user & greet the user.

CODE:

```
<!DOCTYPE html>
<html>
<head>
  <title>Question 3</title>
</head>
<body>

<h2>Greeting Program</h2>

<script>

let userName = prompt("Enter your name:");

document.write("<h3>Hello " + userName + "! Welcome to JavaScript.</h3>");

</script>

</body>
</html>
```

OUTPUT:

Azka

4. Write a program to take input a number from user & display it's multiplication table on your browser. If user does not enter a new number, multiplication table of 5 should be displayed by default.
1. Take
 - a) Take three subjects name from user and store them in 3 different variables.
 - b) Total marks for each subject is 100, store it in another variable.

Take obtained marks for first subject from user and stored it in different variable.

Take obtained marks for remaining 2 subjects from user and store them in variables.

c) Now calculate total marks and percentage and show the result in browser like this.(Hint: user table)



CODE:

```
<!DOCTYPE html>
<html>
<head>
  <title>Question 6</title>
</head>
<body>

  <h2>Marks Sheet Using Table</h2>

  <script>

    let sub1 = prompt("Enter Subject 1");
    let sub2 = prompt("Enter Subject 2");
    let sub3 = prompt("Enter Subject 3");

    let total = 100;

    let m1 = Number(prompt("Marks of " + sub1));
    let m2 = Number(prompt("Marks of " + sub2));
    let m3 = Number(prompt("Marks of " + sub3));

    let totalObtained = m1 + m2 + m3;
    let totalMarks = total * 3;
    let percent = (totalObtained / totalMarks) * 100;

    document.write("<h2>Marks Sheet</h2>");

    document.write("<table border='1' cellpadding='8'>");
    document.write("<tr><th>Subject</th><th>Total Marks</th><th>Obtained Marks</th></tr>");
    document.write("<tr><td>" + sub1 + "</td><td>100</td><td>" + m1 + "</td></tr>");
    document.write("<tr><td>" + sub2 + "</td><td>100</td><td>" + m2 + "</td></tr>");
    document.write("<tr><td>" + sub3 + "</td><td>100</td><td>" + m3 + "</td></tr>");
    document.write("<tr><th>Total</th><th>" + totalMarks + "</th><th>" + totalObtained + "</th></tr>");
    document.write("<tr><th colspan='3'>Percentage = " + percent.toFixed(2) + "%</th></tr>");
    document.write("</table>");

  </script>

</body>
</html>
```

OUTPUT:

Marks Sheet

Subject	Total Marks	Obtained Marks

English	100	80
Math	100	75
Science	100	90

Total	300	245

Percentage = 81.67%